

Book I.

Symbols and Abbreviations

$\therefore$	expresses the word <i>therefore</i> .
$\because$	expresses the word <i>because</i> .
$=$	expresses the word <i>equal</i> . This sign of equality may be read <i>equal to</i> , or <i>is equal to</i> , or <i>are equal to</i> ; but the discrepancy in regard to the introduction of the auxiliary verbs <i>is</i> , <i>are</i> , &c. cannot affect the geometrical rigour.
$\neq$	means the same as if the words ' <i>not equal</i> ' were written.
$>$	signifies <i>greater than</i> .
$<$	signifies <i>less than</i> .
$\nless$	signifies <i>not greater than</i> .
$\nless$	signifies <i>not less than</i> .
$+$	is read <i>plus (more)</i> , the sign of addition; when interposed between two or more magnitudes, signifies their sum.
$-$	is read <i>minus (less)</i> , signifies subtraction; and when placed between two quantities denotes that the latter is taken from the former.

× expresses the product of two or more numbers when placed between them in arithmetic and algebra; but in geometry it is generally used to express a *rectangle*, when placed between “two straight lines which contain one of its right angles.” A *rectangle* may also be represented by placing a point between two of its conterminous sides.

∴ expresses an *analogy* or *proportion*; thus if A, B, C and D represent four magnitudes, and A has to B the same ratio that C has to D, the proportion is written $A : B :: C : D$, $A : B = C : D$, or $\frac{A}{B} = \frac{C}{D}$.

This equality is read: *as A is to B, so is C to D.*

∥ signifies *parallel to*.

⊥ signifies *perpendicular to*.

△ signifies *angle*.

∟ signifies *right angle*.

⌒ signifies *two right angles*.

↑ or briefly designates a *point*.

↯

The square described on a line is concisely written thus, —^2 . In the same manner twice the square of, is expressed by $2 \cdot \text{—}^2$.

def. signifies *definition*.

post. signifies *postulate*.

ax. signifies *axiom*.

hyp. signifies *hypothesis*. The *hypothesis* is the condition assumed or taken for granted. Thus, the hypothesis of the proposition that a triangle is isosceles is that its legs are equal.

const. signifies *construction*. The *construction* is the change made in the original figure, by drawing lines, making angles, describing circles, &c., in order to adapt it to the argument of the demonstration or the solution of the problem. The conditions under which these changes are made are as indisputable as those contained in the hypothesis.

Q. E. D. signifies *Quod erat demonstrandum* — which was to be demonstrated.